

Exercise 3: Stored Procedures

Scenario 1: The bank needs to process monthly interest for all savings accounts.

- **Question:** Write a stored procedure **ProcessMonthlyInterest** that calculates and updates the balance of all savings accounts by applying an interest rate of 1% to the current balance.

Current Accounts Table:

account_id	customer_name	account_type	balance
1	Alice Smith	SAVINGS	1000.00
2	Bob Jones	CHECKING	500.00
3	Charlie Brown	SAVINGS	2500.00

// with cursor

```
CREATE OR REPLACE PROCEDURE processMonthlyInterest IS

    cursor c is select * from Accounts where account_type='SAVINGS' ;
    rec c%rowtype;
begin
    open c;
    loop
        fetch c into rec;
        exit when c%Notfound;
        update Accounts
        set balance = rec.balance +( rec.balance/100)
        where account_id = rec.account_id;

    end loop;
    close c;
    commit;
end ProcessMonthlyInterest ;
/
```

//without cursor

```
CREATE OR REPLACE PROCEDURE processMonthlyInterest IS
BEGIN
    UPDATE Accounts
    SET balance = balance + (balance / 100)
    WHERE account_type = 'SAVINGS';

    COMMIT;
END processMonthlyInterest;
/
```

Scenario 2: The bank wants to implement a bonus scheme for employees based on their performance.

- **Question:** Write a stored procedure **UpdateEmployeeBonus** that updates the salary of employees in a given department by adding a bonus percentage passed as a parameter.

Table used:

Current Employees Table:

emp_id	emp_name	department	salary
1	John Doe	SALES	50000.00
2	Jane Smith	IT	75000.00
3	Emily Davis	SALES	55000.00
4	Michael Brown	HR	60000.00

```
CREATE OR REPLACE PROCEDURE UpdateEmployeeBonus (
    param_department in varchar2,
    param_bonus in number
) IS
BEGIN
    UPDATE Employees
    SET salary = salary + (salary*param_bonus / 100)
    WHERE department = param_department;

    COMMIT;
END UpdateEmployeeBonus;
/
```

Scenario 3: Customers should be able to transfer funds between their accounts.

- **Question:** Write a stored procedure **TransferFunds** that transfers a specified amount from one account to another, checking that the source account has sufficient balance before making the transfer.

Current Accounts Table:

account_id	customer_name	account_type	balance
1	Alice Smith	CHECKING	1500.00
2	Alice Smith	SAVINGS	500.00
3	Bob Jones	CHECKING	200.00

```
create or replace procedure TransferFunds(
    acc1 in number, acc2 in number, money in number
) is
temp number;
BEGIN

    select balance into temp from accounts where account_id = acc1
    if( temp >= money)
    -- select balance into temp2 from accounts where account_id = acc2)
    then
        dbms_output.put_line('have sufficient balance to transfer');
        update accounts
        set balance=balance- money
        where account_id = acc1;

        update accounts
        set balance=balance+money
        where account_id = acc2;

    else dbms_output.put_line('not have sufficient balance to transfer');
    end if;
    commit;

end;
```